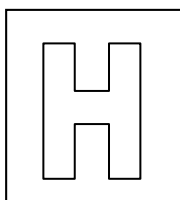


Candidate Name: _____

Class Adm No

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2018 Preliminary Exams Pre-University 3

MATHEMATICS

9758/02

Paper 2

13 Sept 2018

3 hours

Additional Materials: Answer Paper
 List of Formulae (MF26)

READ THESE INSTRUCTIONS FIRST

Write your admission number, name and class on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use an approved graphing calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, arrange your answers in NUMERICAL ORDER and fasten all your work securely together.

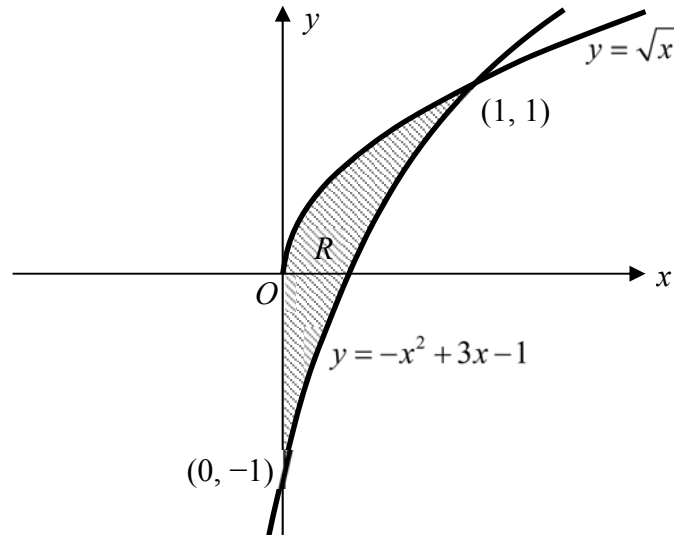
The number of marks is given in brackets [] at the end of each question or part question.

This question paper consists of 8 printed pages.

[Turn over

Section A: Pure Mathematics [40 marks]

- 1** In the diagram, the region R is bounded by the curves $y = -x^2 + 3x - 1$, $y = \sqrt{x}$ and the y -axis.



Without using a graphing calculator, find the volume of the solid generated when R is rotated through 2π radians about the y -axis. [4]

- 2 (a)** A curve C has parametric equations

$$x = k \cos t, \quad y = k \sin 2t, \quad \text{for } 0 \leq t \leq \frac{\pi}{4},$$

where k is a positive constant.

- (i)** Find the equation of the normal to C at the point $(k \cos p, k \sin 2p)$, where p is a constant such that $0 \leq p < \frac{\pi}{4}$. [3]
- (ii)** Deduce the equation of the tangent to C at the point where $t = 0$. [2]
- (b)** A rectangle of length $2|x|$ and breadth $2|y|$ is inscribed in the ellipse $\frac{x^2}{36} + \frac{y^2}{9} = 1$ where (x, y) is any point on the ellipse.
- (i)** Show that the area of the rectangle, A , is given by $A = 8\sqrt{9y^2 - y^4}$. [2]
- (ii)** Find the maximum area of the rectangle. [4]

- 3 (a) Given that $a-i$ is a root of the equation $z^3 + 4(1+i)z^2 + (-2+9i)z - 5+i = 0$, where a is a real constant, show that $(a^3 + 4a^2 + 3a) + (a^2 + a)i = 0$ and find all the roots of the equation. [4]

Deduce the roots of the equation $z^3 + 8(1+i)z^2 + 4(-2+9i)z + 8(-5+i) = 0$. [2]

- (b) It is given that $w = (1-i\sqrt{3})^4$.

(i) Without using a graphing calculator, find the modulus and argument of w . [2]

(ii) Hence find the three smallest positive whole number values of n for which $\frac{w^n}{w^*}$ is a real number. [4]

- 4 Relative to the origin O , the position vectors of points A , B and C are $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively. It is given that $\mathbf{a}$ and $\mathbf{b} - \mathbf{a}$ are perpendicular and C lies on AB produced such that $AC : AB = 4 : 3$.

(i) If $\mathbf{a}$ is a unit vector, show that $|\mathbf{b}| > 1$. [3]

Given further that $|\mathbf{b}| = 2$, find the angle between $\mathbf{a}$ and $\mathbf{b}$. [1]

(ii) The direction cosines of $\mathbf{b}$ are 0.6 , λ , μ and $\mathbf{b}$ is perpendicular to the y -axis. Find

(a) the angle $\mathbf{b}$ makes with the x -axis, [1]

(b) λ and μ . [2]

(iii) By expressing $\mathbf{c}$ in terms of $\mathbf{a}$ and $\mathbf{b}$, show that $|\mathbf{c} \cdot \mathbf{a}| = \mathbf{a} \cdot \mathbf{a}$. [3]

Hence state the length of projection of $\mathbf{c}$ onto $\mathbf{a}$ in terms of $\mathbf{a}$. [1]

(iv) Give a geometrical interpretation of $|\mathbf{c} \times \mathbf{a}|$ and hence evaluate $\frac{|\mathbf{c} \times \mathbf{a}|}{|\mathbf{b} \times \mathbf{a}|}$. [2]

Section B: Probability and Statistics [60 marks]

- 5 A biased die is such that the probability of getting a score of 1, 2, 3, 4, 5 and 6 is $\frac{1}{5}, \frac{1}{6}, \frac{1}{7}, \frac{1}{8}, p$ and q respectively.

(i) If the mean score is 3.5, find p and q . [2]

A game is played by throwing the biased die and a fair coin together. If the coin shows a head, the player wins \$1 more than the score on the die. If the coin shows a tail, the player wins \$1 less than the score on the die. The winnings from one game is denoted by W .

(ii) Find $P(W = w)$ for all possible values of w . [2]

(iii) Without any further calculations, state whether you would play this game if you need to pay \$4 at the start of the game, justifying your answer. [1]

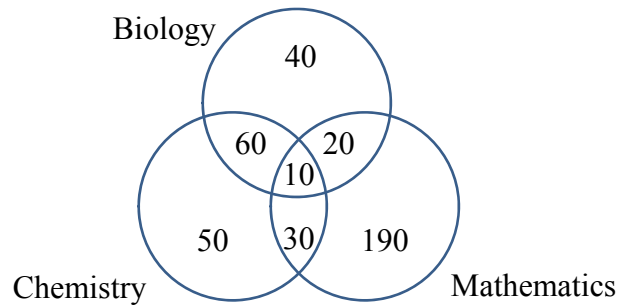
- 6 In the 2018 FIFA World Cup, the Morocco football team started the opening match against Iran with 1 goalkeeper, 3 defenders, 4 midfielders and 3 forwards.

(i) Before the match, the 11 selected players, together with their coach stand in a line for a photo shoot. In how many ways can this be done if the coach and the goalkeeper must stand at either ends and one particular defender and one particular forward must not stand together? [2]

(ii) During half-time, the coach gathered the 11 players at a round table with 12 numbered seats to discuss strategies for the second half of the match. In how many ways can this be done if the 3 defenders must be seated together, the 4 midfielders must be seated together and the 3 forwards must be seated together? [3]

(iii) After the match, the coach decides to construct four-lettered code-words from the 7 letters of the word MOROCCO. How many such code-words are there? [3]

- 7 The Venn diagram below shows the number of students studying Biology, Chemistry and Mathematics in a junior college.



One of the students is chosen at random.

B is the event that the student studies Biology

C is the event that the student studies Chemistry

M is the event that the student studies Mathematics

- (i) Determine if B and C are independent, justifying your answer. [1]
- (ii) Find
- (a) $P(B \cup C')$ and [1]
- (b) $P(C | M \cap B)$. [1]

The junior college also offers Further Mathematics as a subject. A student studying Further Mathematics must also study Mathematics. Given that the event that a randomly chosen student studying Further Mathematics is independent of C , find the largest possible number of students studying Further Mathematics. [3]

- 8 The random variable X has a normal distribution with mean μ and standard deviation 2.

- (i) Given that $P(X \leq 1) = 0.1587$, find the value of μ , giving your answer to the nearest integer. [2]
- (ii) Given further that $P(X > 3k) = P(X < 9k + 4)$ and using the answer obtained in part (i), find the value of k . [1]

The random variable Y is related to X by the formula $Y = 10 - X$.

- (iii) Find $P(\bar{Y} > 6)$, where $\bar{Y}$ is the mean of two independent observations of Y . [3]

[Turn over

- 9 A sample of 10 students are selected from a mixed school to participate in a survey on the school sports facilities. The number of male students in a sample of 10 is denoted by the random variable X and the proportion of male students in the mixed school is p .

- (i) State, in the context of this question, two assumptions required for X to be well modelled by a binomial distribution. [2]

Assume now that these assumptions do in fact hold.

- (ii) Given that $P(X \leq 1) = 0.05$, write down an equation for the value of p , and find this value numerically. [2]

- (iii) 8 such samples of 10 students are selected to participate in the household income survey. Find the probability that exactly 7 of the samples have at least 2 male students. [2]

For the rest of the question, take $p = 0.4$.

- (iv) Find the most probable number of **female** students selected in a sample of 10 students. [2]
- (v) 60 samples of 10 students each are selected to participate in the school climate survey. By using a suitable approximation, estimate the probability that the total number of male students selected exceeds 230. [3]

- 10** The distributor claims that the mean mass of cereal in a randomly chosen packet is 600 grams. A retailer suspects that the mean mass of the cereal is being overstated. He takes a random sample of 50 packets of cereal and weighs the content, x grams, in each packet. The results are summarised as follows:

$$\sum(x - 600) = -8 \text{ and } \sum(x - 600)^2 = 11.3$$

- (i)** Find unbiased estimates of the population mean and variance of the mass of cereal in a packet. [2]
- (ii)** Test, at the 1% level of significance, whether the distributor's claim has been overstated. [4]
- (iii)** State whether there is a need to assume a normal population in conducting the test in part **(ii)**, justifying your answer. [1]
- (iv)** Explain, in this context, the meaning of the p -value of the test obtained in part **(ii)**. [1]

The packaging process has been changed so that the mass of cereal in a randomly chosen packet follows a normal distribution with a standard deviation of 0.5 grams. The distributor now claims that the mean mass of cereal in a randomly chosen packet is μ_0 grams. The retailer selects a new random sample of 25 packets of cereal and the sample mean mass is found to be 600.6 grams. Find the range of possible values of μ_0 so that the retailer's suspicion that the mean mass differs from μ_0 grams is valid at the 5% level of significance. Give your answer correct to 2 decimal places. [4]

[Turn over

- 11** During the National Step Challenge period, participants are able to redeem rewards if they obtained 750 healthpoints in the first tier. In any day during the period, participants will earn 10 healthpoints when they clocked 5000 steps, and 25 healthpoints when they clocked 7500 steps and a maximum of 40 healthpoints when they clocked 10000 steps. Kenny recorded his steps count using his step tracker in the first two weeks as follows:

Day x	1	3	5	7	9	11	13
Steps y (in thousands)	3.8	5.9	6.7	7.5	6.9	8.2	8.5

- (i) Draw a scatter diagram showing the above data. [1]
- (ii) Suggest a possible reason why one of the step counts does not seem to follow the trend. [1]

Assume that the reason given in part (ii) is valid, the outlier is removed.

- (iii) Using the remaining six points, find both equations of the least squares regression line of y on x and that of x on y . [2]
- (iv) Interpret, in the context of this question, the gradient of the least squares regression line of y on x obtained in part (iii). [1]
- (v) Use a suitable regression line found in part (iii) to estimate the number of days taken by Kenny to clock 10000 steps. Justify the choice of the regression line used. [2]

It is decided to fit a model of the form $\ln(L - y) = a + bx$, where L is a suitable constant. The product moment correlation coefficient between x and $\ln(L - y)$ is denoted by r . The following table gives values of r for some possible values of L .

L	10.1	10.2	10.3
r		-0.98424	-0.98318

- (vi) Calculate the value of r for $L=10.1$, giving your answer correct to 5 decimal places. Hence, with the help of the table, suggest with a reason which of 10.1, 10.2 or 10.3 is the most appropriate value for L . [2]
- (vii) Using this value of L , calculate the values of a and b , and use them to predict the number of steps clocked by Kenny on Day 10. Comment on the reliability of this estimate. [3]

END OF PAPER